SE LAB 6
PASSWORD POLICY

CODE COVERAGE TEST CASES:

```
from password_policy import check_password, strength, suggest_fix


def test_valid_password():
    assert check_password("Abcdef1!") == []


def test_password_length():
    assert "too short" in check_password("abc")
    assert "too long" in check_password("A" + "a" * 63 + "1!")


def test_common_password():
    assert "common" in check_password("password")


def test_missing_character_types():
    assert "no uppercase" in check_password("abcdefg1!")
    assert "no lowercase" in check_password("ABCDEFG1!")
    assert "no digit" in check_password("Abcdefgh!")
    assert "no special" in check_password("Abcdefg1")


def test_password_strength():
    assert strength("Abcdefghijk1!Z") == "strong"
    assert strength("Abcdef1!") == "good"
    assert strength("abcdefg1!") == "weak"
    assert strength("abc") == "very weak"


def test_suggest_fix():
    assert suggest_fix([]) == "none needed"
    assert suggest_fix(["too short"]) == "add more characters"
    assert suggest_fix(["common"]) == "avoid common passwords"
    assert suggest_fix(["no uppercase"]) == "add uppercase"
    assert suggest_fix(["too long"]) == "shorten it"
```

```
C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6>python -m
 coverage run --source=password_policy -m pytest CodeCoverageTest_password_p
olicy.py -v
========================== test session starts ==========================
platform win32 -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Python315\
python.exe
cachedir: .pytest_cache
rootdir: C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6
collected 6 items

CodeCoverageTest_password_policy.py::test_valid_password PASSED      [ 16%]
CodeCoverageTest_password_policy.py::test_password_length PASSED     [ 33%]
CodeCoverageTest_password_policy.py::test_common_password PASSED     [ 50%]
CodeCoverageTest_password_policy.py::test_missing_character_types PASSED [ 6
6%]
CodeCoverageTest_password_policy.py::test_password_strength PASSED   [ 83%]
CodeCoverageTest_password_policy.py::test_suggest_fix PASSED         [100%]

=========================== 6 passed in 0.10s ===========================

C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6>python -m
 coverage report -m
Name                 Stmts   Miss  Cover   Missing
--------------------------------------------------
password_policy.py      47      0   100%
--------------------------------------------------
TOTAL                   47      0   100%
```

BRANCH COVERAGE TEST CASES:

```
from password_policy import check_password, strength, suggest_fix


def test_all_password_character_branches():
    assert check_password("Abcdef1!") == []
    assert check_password("abcdefg1!") == ["no uppercase"]
    assert check_password("ABCDEFG1!") == ["no lowercase"]
    assert check_password("Abcdefgh!") == ["no digit"]
    assert check_password("Abcdefg1") == ["no special"]


def test_length_branches():
    assert "too short" in check_password("abc")
    assert "too long" in check_password("A" + "a" * 63 + "1!")
    assert check_password("Abcdef1!") == []
```

```
def test_common_password_branches():
    assert "common" in check_password("password")
    assert "common" not in check_password("Abcdef1!")


def test_strength_branches():
    assert strength("Abcdefghijk1!Z") == "strong"
    assert strength("Abcdef1!") == "good"
    assert strength("abcdefg1!") == "weak"
    assert strength("abc") == "very weak"


def test_suggest_fix_branches():
    assert suggest_fix([]) == "none needed"
    assert suggest_fix(["too short"]) == "add more characters"
    assert suggest_fix(["common"]) == "avoid common passwords"
    assert suggest_fix(["no uppercase"]) == "add uppercase"
    assert suggest_fix(["too long"]) == "shorten it"
```

```
C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6>python -m coverage r
un --branch --source=password_policy -m pytest BranchCoverageTest_password_policy.py -v
========================= test session starts =========================
platform win32 -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Python315\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6
collected 5 items

BranchCoverageTest_password_policy.py::test_all_password_character_branches PASSED [ 20
%]
BranchCoverageTest_password_policy.py::test_length_branches PASSED [ 40%]
BranchCoverageTest_password_policy.py::test_common_password_branches PASSED [ 60%]
BranchCoverageTest_password_policy.py::test_strength_branches PASSED [ 80%]
BranchCoverageTest_password_policy.py::test_suggest_fix_branches PASSED [100%]

========================= 5 passed in 0.13s =========================

C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6>python -m coverage r
eport -m
Name                 Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------
password_policy.py      47      0     36      0   100%
----------------------------------------------------------------
TOTAL                   47      0     36      0   100%

C:\Users\BELA KAPOOR\OneDrive\Desktop\SE_LAB6_password_policy\lab6>git add CodeCoverage
Test_password_policy.py 01_Code_Coverage.png
fatal: pathspec '01_Code_Coverage.png' did not match any files
```